



BITS Pilani

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CE F231 Fluid Mechanics

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Lecture 13: Fluid Kinematics

Learning Outcome

- Methods of describing fluid motion
- Types of fluid flow
- Velocity and acceleration
- Streamlines, path lines, and streak lines
- Control mass, control volume, and continuity equation
- Velocity potential and stream functions
- Types of motion
- Vortex flow

Methods of Describing Fluid Motion

➤ Lagrangian Approach

- Individual fluid particles are followed during its motion, and displacement, velocity, and acceleration are described.

➤ Eulerian Approach

- Displacement, velocity, and acceleration are described at specific points in the flow field

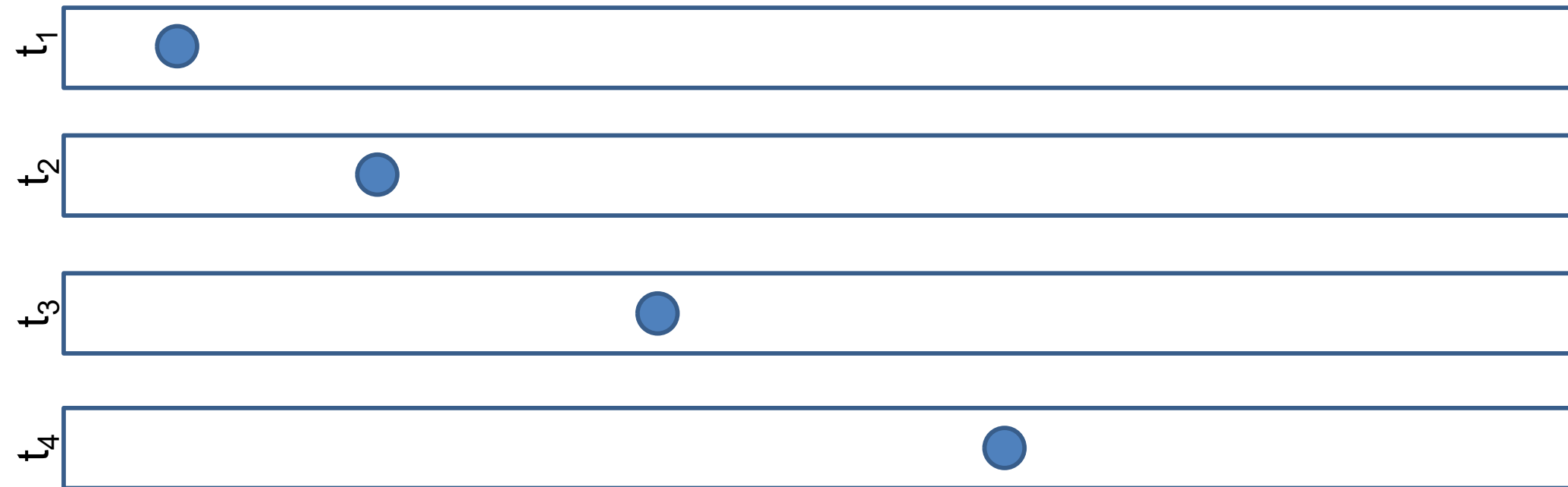


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Methods of Describing Fluid Motion

➤ Lagrangian Approach

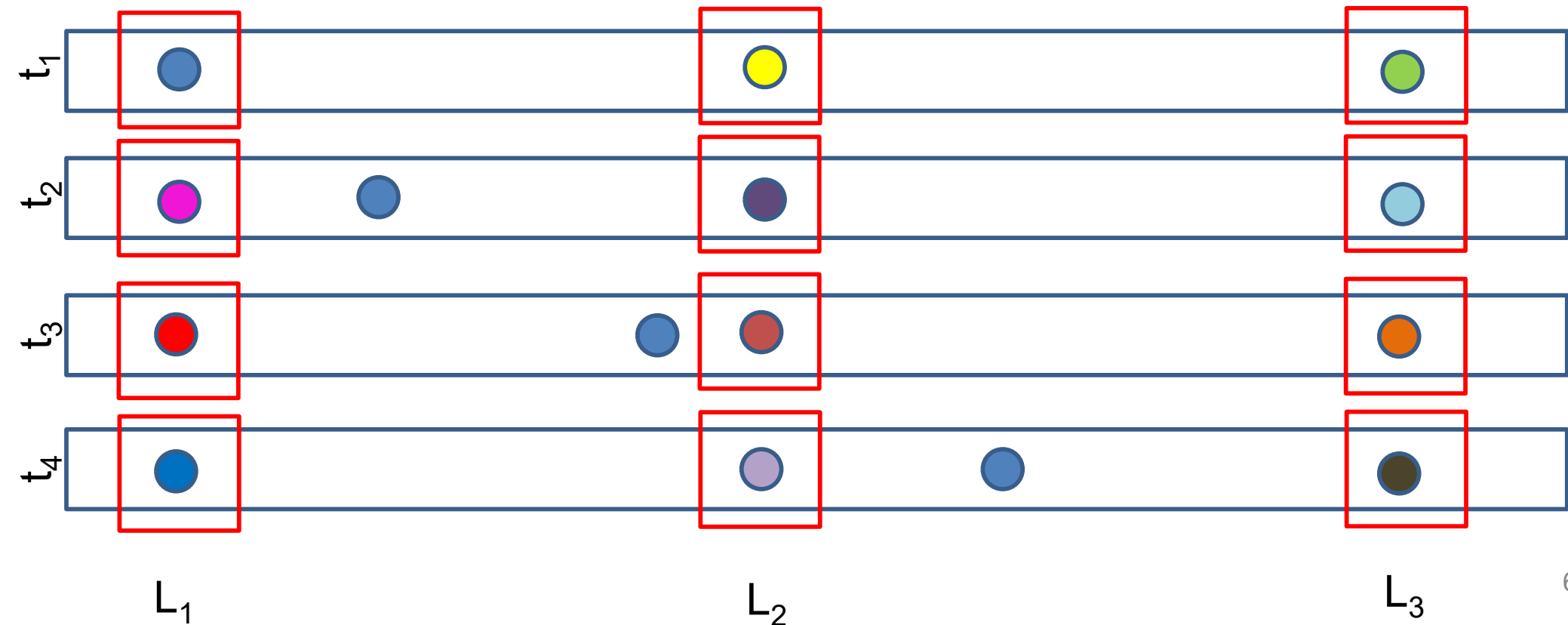
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Methods of Describing Fluid Motion

➤ Eulerian Approach

- Displacement, velocity, and acceleration are described at specific points in the flow field



Methods of Describing Fluid Motion

➤ Lagrangian Approach



Methods of Describing Fluid Motion

➤ Eulerian Approach



Types of Fluid Flow

➤ Steady Flow

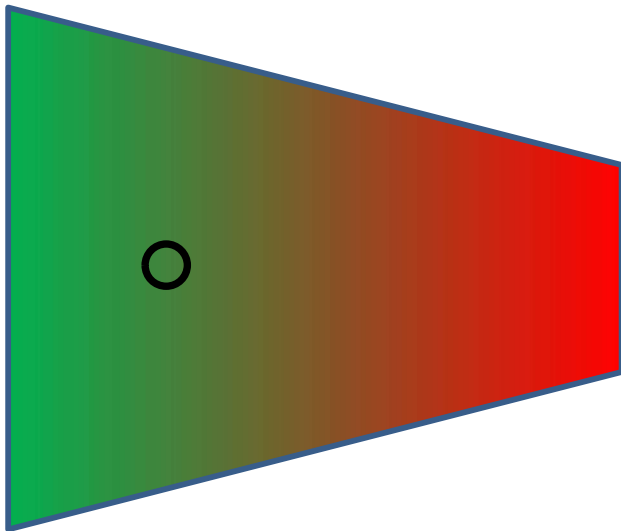
- Steady Flow: Fluid characteristics at any point in fluid flow do not change with time

10 m/s 20 m/s 30 m/s

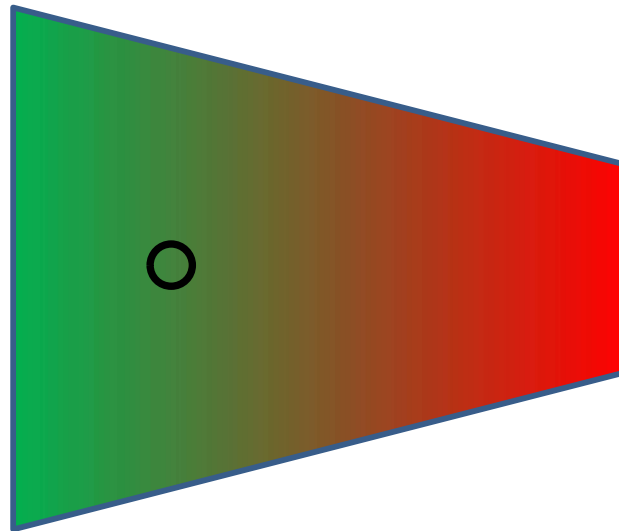


$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial \rho}{\partial t}\right)_{x_0, y_0, z_0} = 0$$

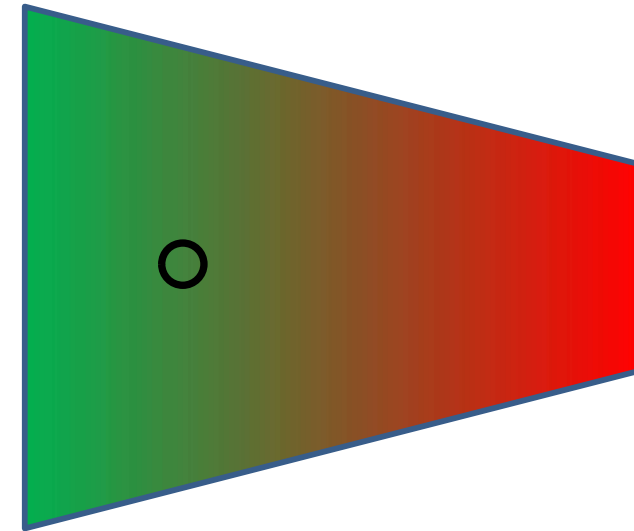
where (x_0, y_0, z_0) is a fixed point in fluid field.



At time $t = t_1$



At time $t = t_2$

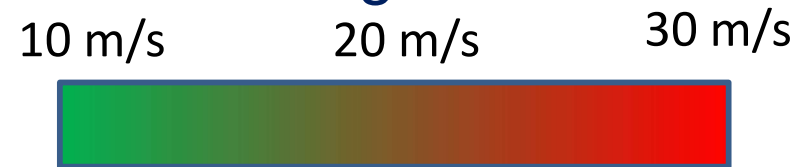


At time $t = t_3$

Types of Fluid Flow

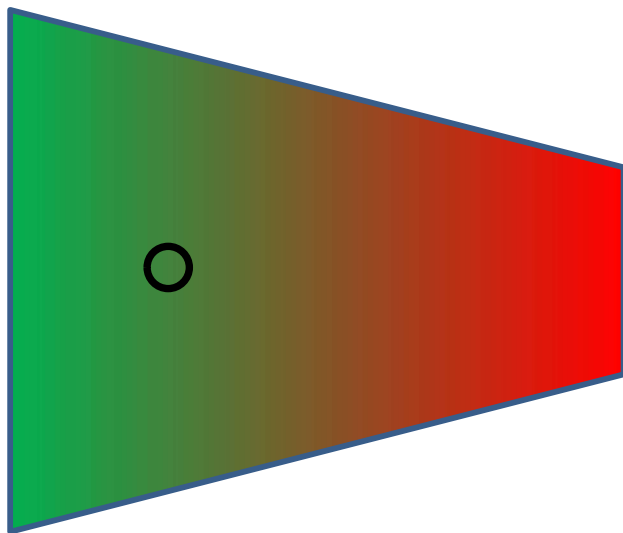
➤ Unsteady Flow

- Unsteady Flow: Fluid characteristics at any point in fluid flow change with time

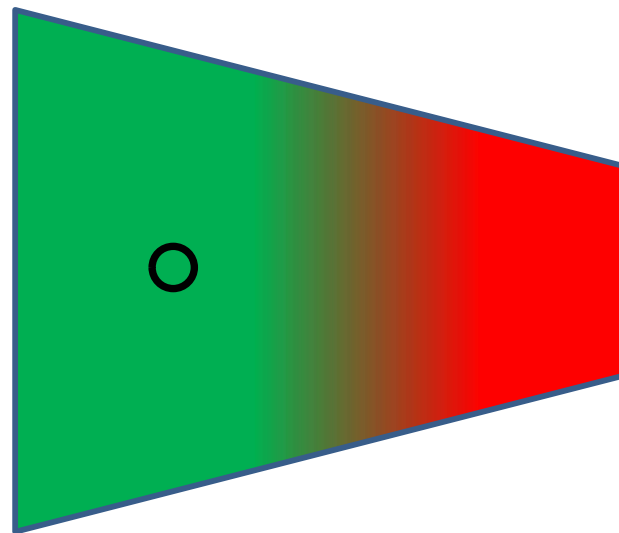


$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} \neq 0, \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} \neq 0 \text{ etc.}$$

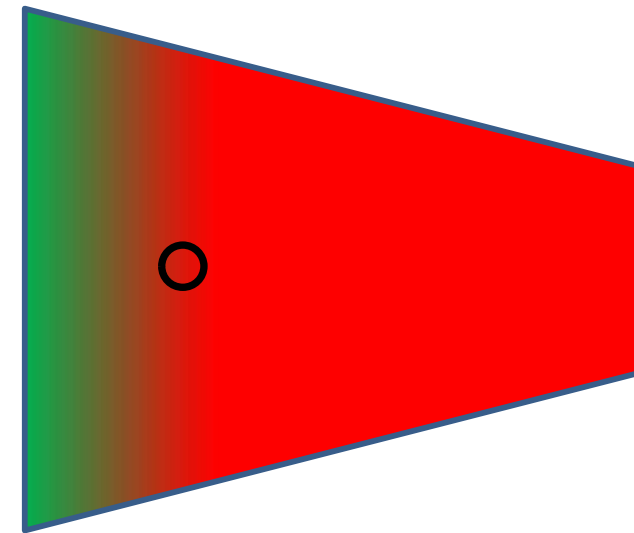
where (x_0, y_0, z_0) is a fixed point in fluid field.



At time $t = t_1$



At time $t = t_2$



At time $t = t_3$

Types of Fluid Flow

➤ Uniform Flow

- Uniform Flow: Fluid characteristics at any time do not change with space

$$\left(\frac{\partial V}{\partial s} \right)_{t = \text{constant}} = 0$$

where ∂V = Change of velocity
 ∂s = Length of flow in the direction S .

10 m/s 20 m/s 30 m/s



At time $t = t_1$



At time $t = t_2$



At time $t = t_3$

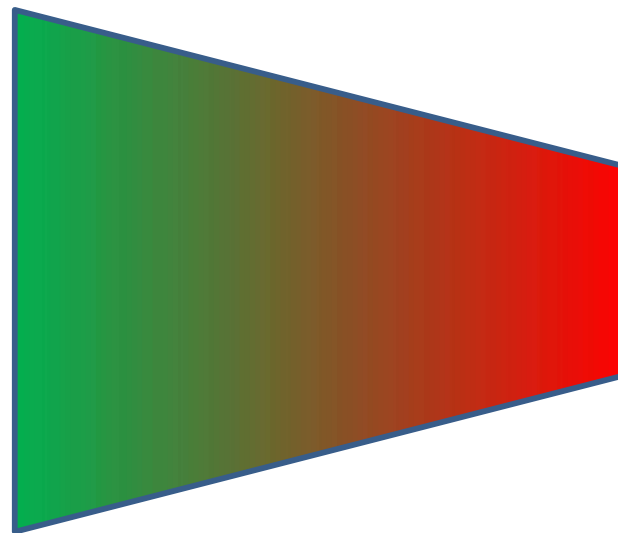
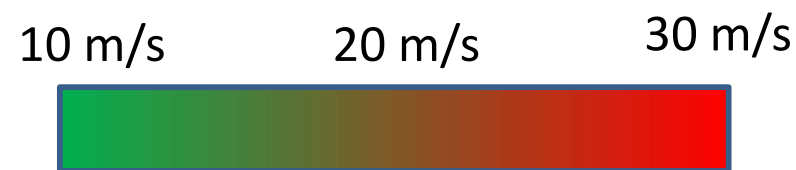
Types of Fluid Flow

➤ Non-uniform Flow

- Non-uniform Flow: Fluid characteristics at any given time changes with space

$$\left(\frac{\partial V}{\partial s} \right)_{t = \text{constant}} \neq 0.$$

where ∂V = Change of velocity
 ∂s = Length of flow in the direction S .

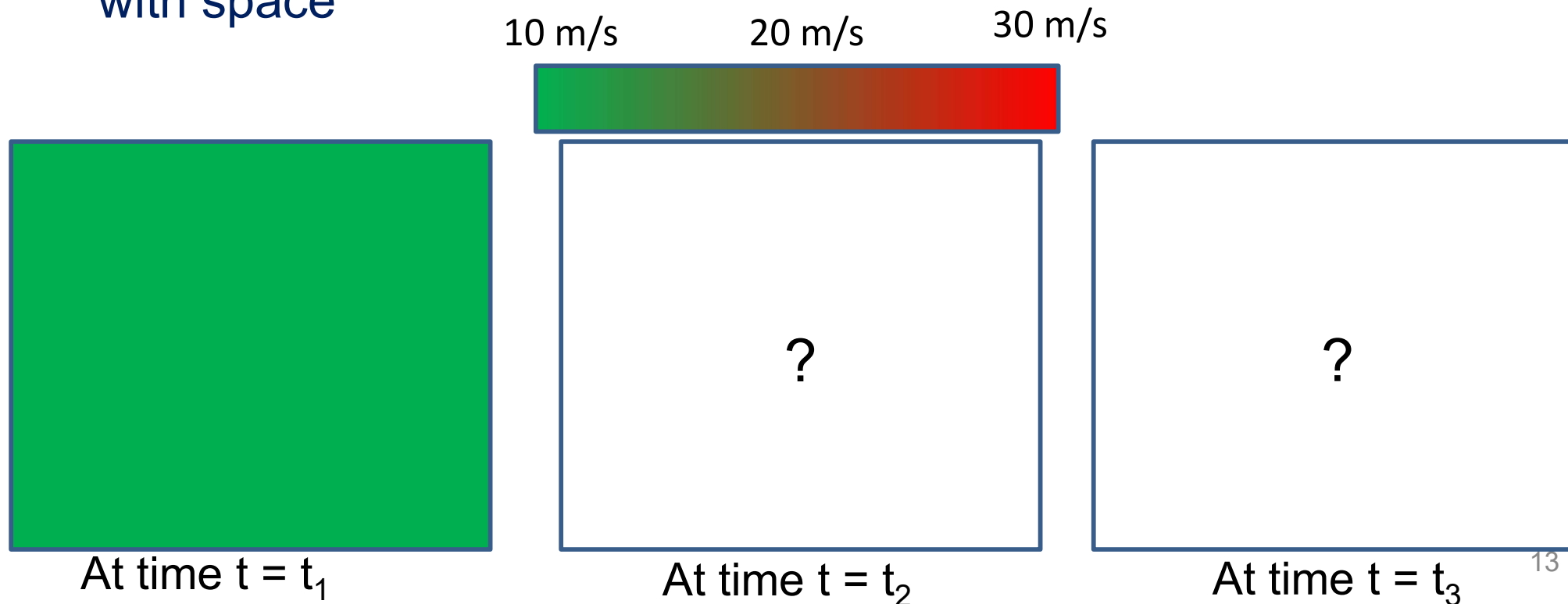


At time $t = t_1$

Question

➤ Steady and Uniform Flow

- Steady Flow: Fluid characteristics at any point in fluid flow do not change with time
- Uniform Flow: Fluid characteristics at any time do not change with space



Types of Fluid Flow

➤ Laminar Flow

- Laminar Flow: Fluid particles move along well-defined paths that are straight and parallel.
- In laminar flow, fluid particles move in layers (laminae) gliding over adjacent layers
- Reynolds number
(Re) = inertial
force/viscous force
- $Re = (\rho V L) / \mu$
- $Re < 2000$

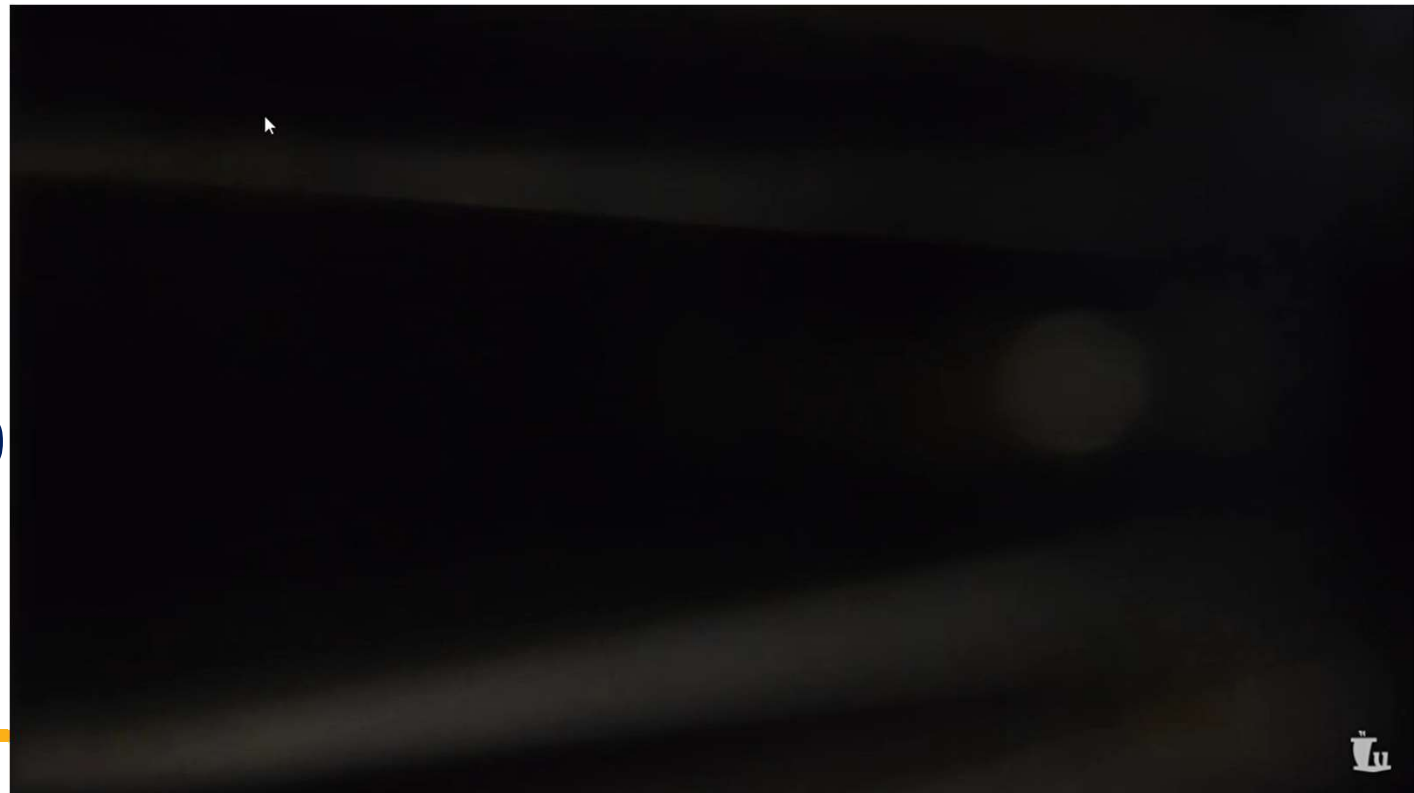
○ Laminar



Types of Fluid Flow

➤ Turbulent Flow

- Turbulent Flow: Fluid particles move in a zig-zag way
- Eddies are formed → High energy loss
- Reynolds number (Re) = inertial force/viscous force
- $Re = (\rho V L) / \mu$
- $Re < 2000$
 - Laminar
- $Re > 4000$
 - Turbulent
- $2000 < Re < 4000$
 - Transitional



Types of Fluid Flow

➤ Rotational and Irrotational Flow

- Rotational flow
 - Flow in which fluid particles rotate about their axis while flowing
- Irrotational flow
 - Fluid particles do not rotate about their own axis while flowing



Types of Fluid Flow

➤ Compressible and Incompressible Flow

■ Compressible flow

- Flow in which density of fluid changes from one point to another
- Density is not constant for the fluid
- Gases are compressible

■ Incompressible flow

- Flow in which density of fluid does not change from one point to another
- Liquids are generally considered as incompressible

Types of Fluid Flow

➤ 1D, 2D and 3D Flow

■ 1D flow

- Flow parameters are functions of one space coordinate only

$$u = f(x), v = 0 \text{ and } w = 0$$

■ 2D flow

- Flow parameters are functions of two space coordinates

$$u = f_1(x, y), v = f_2(x, y) \text{ and } w = 0.$$

■ 3D flow

- Flow parameters are functions of all three space coordinates

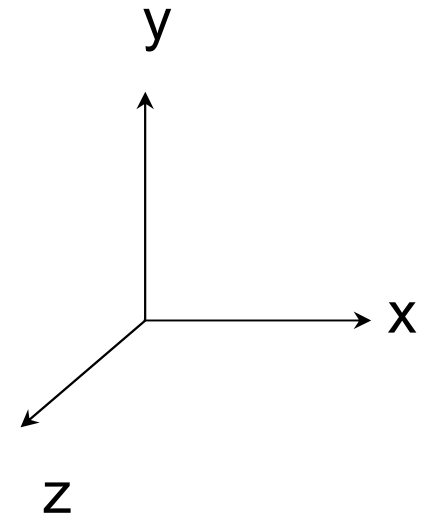
$$u = f_1(x, y, z), v = f_2(x, y, z) \text{ and } w = f_3(x, y, z).$$

Velocity

x-component of velocity $u = f_1(x, y, z, t)$

y-component of velocity $v = f_2(x, y, z, t)$

z-component of velocity $w = f_3(x, y, z, t)$



Resultant velocity, $V = ui + vj + wk$

Velocity Field



Velocity Field (Experimental – PIV)



Velocity Field



Velocity Field (Experimental – PIV)

